

Math 147

Analysis

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Contents

1	Course Notes	3
1.1	General Information	3
1.1.1	Contact	3
1.1.2	Course Structure	3
2	Foundations of Calculus	4
2.1	Lecture 1	4
2.1.1	What Is Calculus?	4
2.1.2	Numbers	4
2.1.3	Statements and Quantifiers	5
2.2	Mathematical Induction	6
2.2.1	Well-Ordering Principle	8
2.3	The Algebraic and Order Properties of $\mathbb{R}$	10
2.3.1	The Absolute Value	14
2.4	Bounds	17
2.5	The Completeness Property of $\mathbb{R}$	19
2.5.1	The Archimedean Property	21
2.5.2	Density of $\mathbb{Q}$	25
3	Sequences and Limits	29
3.1	Limits of Sequences	29
3.1.1	Divergence	30

1 Course Notes

1.1 General Information

1.1.1 Contact

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Office hours: Monday, 3:30–4:30 p.m., and Friday, 10:00–11:00 a.m., in MC 4006.

1.1.2 Course Structure

Lectures: Section 002 meets on Monday, Wednesday, and Friday from 12:30–1:20 p.m. in MC 4021.

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2 Foundations of Calculus

2.1 Lecture 1

2.1.1 What Is Calculus?

Differential and integral calculus are two branches of calculus.

Differential calculus: The study of slopes and tangent lines.

Integral calculus: The study of areas and accumulation.

2.1.2 Numbers

The number systems discussed in this lecture include

$$\begin{aligned}\mathbb{N} &= \{1, 2, 3, \dots\}, \\ \mathbb{Z} &= \{\dots, -1, 0, 1, \dots\}, \\ \mathbb{Q} &= \left\{ \frac{m}{n} : m, n \in \mathbb{Z}, n \neq 0 \right\}.\end{aligned}$$

The natural numbers are closed under addition. The integers are closed under addition, subtraction, and multiplication, but they are not closed under division; for example, $1/2 \notin \mathbb{Z}$. Therefore, we introduce the rational numbers to include quotients of integers with nonzero denominators.

Theorem 2.1.1 (Irrationality of $\sqrt{2}$)

The number $\sqrt{2}$ is irrational; equivalently, $\sqrt{2} \notin \mathbb{Q}$.

Remark (Proof idea)

We argue by contradiction. If $\sqrt{2}$ were rational, we could write it as a fraction m/n in *lowest terms*, i.e. with $\gcd(m, n) = 1$. Squaring turns the equation into a statement about parity: $m^2 = 2n^2$ forces m to be even, and feeding this back into the equation forces n to be even as well. But then m and n share the factor 2, contradicting the assumption that they had no common factor. The key mechanism is that “lowest terms” gives us a parity constraint that the equation $m^2 = 2n^2$ cannot simultaneously satisfy.

Proof

Suppose, for contradiction, that $\sqrt{2} \in \mathbb{Q}$. Then we can write

$$\sqrt{2} = \frac{m}{n}, \quad m, n \in \mathbb{Z}, \quad n > 0, \quad \gcd(m, n) = 1.$$

Squaring both sides gives

$$2 = \frac{m^2}{n^2} \implies m^2 = 2n^2.$$

Thus m^2 is even. An odd integer has an odd square, so m must be even.

Write $m = 2\ell$ for some $\ell \in \mathbb{Z}$. Substituting this into the equation above gives

$$\begin{aligned} 2n^2 &= 4\ell^2, \\ n^2 &= 2\ell^2. \end{aligned}$$

Therefore n^2 is even, and the same parity argument shows that n is even.

This means that both m and n are even, contradicting $\gcd(m, n) = 1$. Therefore, $\sqrt{2} \notin \mathbb{Q}$. ■

Assuming lowest terms turns the equation into a parity contradiction.

The irrational numbers are the real numbers that are not rational:

$$\mathbb{R} \setminus \mathbb{Q} = \{x \in \mathbb{R} : x \notin \mathbb{Q}\}.$$

Example 2.1.2 (Decimal representation)

Every nonnegative real number has a decimal representation of the form

$$x = a + \sum_{i=1}^{\infty} \frac{x_i}{10^i}, \quad a \in \mathbb{N} \cup \{0\}, \quad x_i \in \{0, 1, 2, \dots, 9\}.$$

Negative real numbers are the negatives of such expansions. Some numbers have two decimal representations, for example $1.999\cdots = 2.000\cdots$.

Example 2.1.3 (Recurring decimal)

Is the statement $1.999\cdots = 2$ correct?

Yes. Let $x = 1.999\cdots$. Then

$$\begin{aligned} x &= 1.999\cdots, \\ 10x &= 19.999\cdots, \\ 9x &= 18, \\ x &= 2. \end{aligned}$$

2.1.3 Statements and Quantifiers

Definition 2.1.4 (Statement)

A statement is a sentence that has a definite truth value: it is either true or false.

The **universal quantifier** $\forall$ means “for all.” For example,

Example 2.1.5

The statement

$$\forall x \in \mathbb{R}, \quad x^2 - x \geq 0$$

is false; for instance, it fails when $x = \frac{1}{2}$.

The **existential quantifier** $\exists$ means “there exists.” For example,

Example 2.1.6

The statement

$$\exists x \in \mathbb{R}, \quad x^2 - x \geq 0$$

is true; for instance, $x = 2$ satisfies the inequality.

Proposition 2.1.7

For every $x \in \mathbb{R}$, there exists $y \in \mathbb{R}$ such that

$$x^3 - y^3 = 1.$$

Proof

Let $x \in \mathbb{R}$ and take $y = \sqrt[3]{x^3 - 1}$. Then

$$x^3 - y^3 = x^3 - (x^3 - 1) = 1.$$

■

Here the choice of y is allowed to depend on x : for each x we produce a (possibly different) y that makes the equation hold. This is exactly what the quantifier order $\forall x \exists y$ permits.

Proposition 2.1.8

The statement

$$\exists y \in \mathbb{R}, \quad \forall x \in \mathbb{R}, \quad x^3 - y^3 = 1$$

is false.

Its negation is

$$\forall y \in \mathbb{R}, \quad \exists x \in \mathbb{R}, \quad x^3 - y^3 \neq 1.$$

To prove the negation, let $y \in \mathbb{R}$ and take $x = y$. Then $x^3 - y^3 = 0 \neq 1$.

Remark

In this second statement, the quantifier order is reversed: $\exists y \forall x$ demands a *single* y that works simultaneously for every $x \in \mathbb{R}$. This is a much stronger requirement than $\forall x \exists y$, which only asks that a (possibly different) y exist for each individual x . The first proposition shows such y 's exist, but they genuinely depend on x (namely $y = \sqrt[3]{x^3 - 1}$), so no single choice of y can work for all x at once. This is why swapping the order of $\forall$ and $\exists$ can change a true statement into a false one, and it is a caution to always read quantified statements in the order they are written.

2.2 Mathematical Induction

Definition 2.2.1 (Principle of Mathematical Induction)

Let $P(n)$ be a statement depending on $n \in \mathbb{N}$. If the following two statements are true:

- (1) $P(1)$ is true;
- (2) for every $k \in \mathbb{N}$, if $P(k)$ is true, then $P(k+1)$ is true;

then

$$\forall n \in \mathbb{N}, \quad P(n)$$

is true.

Remark (How to use induction in practice)

To prove that $P(n)$ holds for every $n \in \mathbb{N}$ by induction, it suffices to carry out two steps: first, verify the *base case* $P(1)$ directly; second, carry out the *inductive step* by fixing an arbitrary $k \in \mathbb{N}$, assuming $P(k)$ holds (the *inductive hypothesis*), and using this assumption to deduce $P(k+1)$. Once both steps are complete, the principle of mathematical induction guarantees $P(n)$ for all $n \in \mathbb{N}$ — no further verification of individual cases is needed.

Here is an example:

Proposition 2.2.2 (Sum of the First n Natural Numbers)

For every $n \in \mathbb{N}$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof

Define $P(n)$ to be the statement

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

The base case is $n = 1$:

$$1 = \frac{1(1+1)}{2}.$$

For the inductive step, let $k \in \mathbb{N}$ and assume that $P(k)$ is true. Then

$$\begin{aligned} 1 + 2 + \cdots + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

Thus, $P(k+1)$ is true. By the principle of mathematical induction, the proposition follows. ■

The principle of mathematical induction has the following equivalent set formulation.

Proposition 2.2.3

Let $S \subseteq \mathbb{N}$. If

- (i) $1 \in S$;
- (ii) for every $k \in S$, $k + 1 \in S$;

then $S = \mathbb{N}$.

Proof

Define $P(n)$ to be the statement $n \in S$. The two assumptions give the base case $P(1)$ and the inductive implication $P(k) \Rightarrow P(k + 1)$. Hence, by mathematical induction, every $n \in \mathbb{N}$ belongs to S . Since $S \subseteq \mathbb{N}$, it follows that $S = \mathbb{N}$. ■

2.2.1 Well-Ordering Principle**Theorem 2.2.4** (Well-Ordering Principle)

Every nonempty subset S of $\mathbb{N}$ has a smallest element.

Remark (Proof idea)

We argue by contradiction: suppose S has no smallest element. We want to show this forces $S = \emptyset$, contradicting the hypothesis that S is nonempty. The natural tool is induction, so we build a set T designed to capture, for each n , whether S has been "avoided" up through n ; showing $T = \mathbb{N}$ by induction will then show S is empty.

Proof

Let $S \subseteq \mathbb{N}$ be nonempty. Suppose, for contradiction, that S has no smallest element. Define

$$T = \{n \in \mathbb{N} : \{1, 2, \dots, n\} \cap S = \emptyset\}.$$

That is, T consists of exactly those $n \in \mathbb{N}$ for which none of $1, 2, \dots, n$ lies in S — informally, $n \in T$ means S has "not yet been seen" by the time we reach n . If we can show $T = \mathbb{N}$, then in particular no natural number lies in S (since every $n \in \mathbb{N}$ would satisfy $n \in T$, forcing $n \notin S$), giving $S = \emptyset$ and the desired contradiction. We show $T = \mathbb{N}$ by induction on n .

Base case. Since S has no smallest element and 1 is the smallest element of $\mathbb{N}$, we must have $1 \notin S$ (otherwise 1 would be the smallest element of S). Hence $\{1\} \cap S = \emptyset$, so $1 \in T$.

Inductive step. Suppose $k \in T$, i.e., $\{1, 2, \dots, k\} \cap S = \emptyset$. We show $k + 1 \in T$. Suppose, for contradiction, that $k + 1 \notin T$; then $\{1, 2, \dots, k + 1\} \cap S \neq \emptyset$. Since $\{1, \dots, k\} \cap S = \emptyset$ by the inductive hypothesis, the only way for $\{1, \dots, k + 1\} \cap S$ to be nonempty is if $k + 1 \in S$. But then $k + 1 \in S$ while none of $1, 2, \dots, k$ lie in S , so $k + 1$ would be the smallest element of S — contradicting our assumption that S has no smallest element. Hence $k + 1 \in T$.

By mathematical induction, $T = \mathbb{N}$. This implies that no natural number belongs to S , so $S = \emptyset$, contradicting the assumption that S is nonempty.

Therefore, S has a smallest element. ■

Remark (Practical use of well-ordering)

In practice, the well-ordering principle is most often invoked as follows: to show a property $P(n)$ holds for every $n \in \mathbb{N}$, argue by contradiction. If $P(n)$ failed for some n , the set of counterexamples $\{n \in \mathbb{N} : \neg P(n)\}$ would be a nonempty subset of $\mathbb{N}$, and the well-ordering principle would guarantee it has a least element k . One then derives a contradiction using the minimality of k — typically by showing that $P(k-1)$ must hold (since $k-1$ is smaller than k and hence not a counterexample) and that $P(k-1)$ forces $P(k)$, contradicting k being a counterexample. This is exactly the strategy used in the proposition below.

Proposition 2.2.5

Using the well-ordering principle, prove that for every $n \in \mathbb{N}$,

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Remark (Proof idea)

We prove the claim by minimal counterexample. Suppose the formula failed for some n ; collect all such failures into a set $S = \{n \in \mathbb{N} : \neg P(n)\}$ — this is the natural set to consider because if we can rule out S being nonempty, then $P(n)$ holds for every n . Since $S \subseteq \mathbb{N}$, the well-ordering principle (rather than induction directly) is the right tool to extract a concrete least counterexample $k = \min S$. We then show $P(k-1) \Rightarrow P(k)$ by direct computation, so k cannot actually be a counterexample — a contradiction.

Remark (Why min, not sup)

It is worth pausing on why we take the *minimum* of S here rather than some supremum-type argument. The well-ordering principle applies specifically to nonempty subsets of $\mathbb{N}$ (not to general subsets of $\mathbb{R}$), and it guarantees the existence of a smallest element that actually *belongs to* S — this is what allows us to write $k = \min S \in S$ and conclude $\neg P(k)$. This is a stronger statement than merely having an infimum: for a general subset of $\mathbb{R}$, an infimum need not belong to the set, so we could not conclude $\neg P(\inf S)$. Here, because $S \subseteq \mathbb{N}$ and S is nonempty, well-ordering hands us an actual least element k with $\neg P(k)$ true, which is exactly what is needed to test P at k and at $k-1$.

It is also worth noting that induction and the well-ordering principle are logically equivalent ways of expressing the same underlying fact about $\mathbb{N}$: the proof of the well-ordering principle above used induction, and here we use well-ordering to reprove a statement ($P(n)$ for all n) that could equally well have been proved by direct induction on n . The two techniques are simply different packagings of the same idea.

Proof

Let $P(n)$ denote the statement

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Suppose, for contradiction, that $P(n)$ fails for some n ; that is,

$$\neg(\forall n \in \mathbb{N}, P(n)) \quad \text{equivalently} \quad \exists n \in \mathbb{N}, \neg P(n).$$

Then the set

$$S := \{n \in \mathbb{N} : \neg P(n)\}$$

is nonempty. By the well-ordering principle, S has a smallest element k . Since $P(1)$ holds, $1 \notin S$, so $k > 1$ and hence $k-1 \in \mathbb{N}$.

By minimality of k , we have $k-1 \notin S$, so $P(k-1)$ is true:

$$1^2 + 2^2 + \cdots + (k-1)^2 = \frac{(k-1)k(2(k-1)+1)}{6} = \frac{(k-1)k(2k-1)}{6}.$$

Adding k^2 to both sides gives

$$\begin{aligned} 1^2 + \cdots + (k-1)^2 + k^2 &= \frac{(k-1)k(2k-1)}{6} + k^2 \\ &= \frac{(k-1)k(2k-1) + 6k^2}{6} \\ &= \frac{k((k-1)(2k-1) + 6k)}{6} \\ &= \frac{k(2k^2 + 3k + 1)}{6} = \frac{k(k+1)(2k+1)}{6}, \end{aligned}$$

so $P(k)$ is in fact true. This contradicts $k \in S$. Therefore $S = \emptyset$, and $P(n)$ holds for every $n \in \mathbb{N}$. ■

Remark (General pattern: minimal counterexample via well-ordering)

To show $P(n)$ holds for all $n \in \mathbb{N}$: assume the counterexample set $S = \{n : \neg P(n)\}$ is nonempty, let $k = \min S$ (which exists by well-ordering), show $P(k-1)$ holds by minimality of k , then show $P(k-1) \Rightarrow P(k)$ — contradicting $k \in S$.

2.3 The Algebraic and Order Properties of $\mathbb{R}$

The real numbers $\mathbb{R}$ form a *complete ordered field*. We build up this description in three stages: first the field (algebraic) axioms, which govern $+$ and $\cdot$; then the order axioms, which let us compare elements with $<$; and finally the completeness axiom, which is what distinguishes $\mathbb{R}$ from $\mathbb{Q}$ and is the source of all limiting behaviour in analysis. Each stage adds structure on top of the previous one, and each is needed for a different purpose: the field axioms let us compute, the order axioms let us compare, and completeness lets us take limits.

Definition 2.3.1 (Field)

A **field** is a set F equipped with two binary operations

$$+ : F \times F \rightarrow F, \quad \cdot : F \times F \rightarrow F,$$

satisfying the following axioms for all $a, b, c \in F$.

F1. Associativity of addition.

$$a + (b + c) = (a + b) + c.$$

F2. Commutativity of addition.

$$a + b = b + a.$$

F3. Additive identity. There exists an element $0 \in F$ such that

$$a + 0 = a.$$

F4. Additive inverses. For every $a \in F$, there exists an element $-a \in F$ such that

$$a + (-a) = 0.$$

F5. Associativity of multiplication.

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c.$$

F6. Commutativity of multiplication.

$$a \cdot b = b \cdot a.$$

F7. Multiplicative identity. There exists an element $1 \in F$, with $1 \neq 0$, such that

$$a \cdot 1 = a.$$

F8. Multiplicative inverses. For every $a \in F$ with $a \neq 0$, there exists an element $a^{-1} \in F$ such that

$$a \cdot a^{-1} = 1.$$

F9. Distributive law.

$$a \cdot (b + c) = a \cdot b + a \cdot c.$$

Remark

The rational numbers $\mathbb{Q}$ also form a field: all nine axioms F1–F9 hold for the usual $+$ and $\cdot$ on fractions. So being a field alone does not distinguish $\mathbb{R}$ from $\mathbb{Q}$; the field axioms only capture the algebra of $+$ and $\cdot$, not any notion of size, order, or completeness.

Remark

In practice, F1–F9 are the axioms that justify every "obvious" algebraic manipu-

lation: moving terms across an equals sign uses additive inverses and associativity, cancelling a nonzero factor uses multiplicative inverses, and expanding a product uses the distributive law. The theorem below derives a few of the most basic consequences (uniqueness of the identities, and $a \cdot 0 = 0$) directly from the axioms, to illustrate that even these familiar facts are theorems, not axioms, once the field axioms are fixed.

Theorem 2.3.2 a) If $z, a \in \mathbb{R}$ with $z + a = a$, then $z = 0$.
 b) If $u \in \mathbb{R}$ and $b \neq 0$ with $u \cdot b = b$, then $u = 1$.
 c) If $a \in \mathbb{R}$, then $a \cdot 0 = 0$.

Proof

a)

$$z = z + 0 = z + (a - a) = (z + a) + (-a) = a + (-a) = 0.$$

b) Multiplying $u \cdot b = b$ by b^{-1} ,

$$u = u \cdot 1 = u \cdot (b \cdot b^{-1}) = (u \cdot b) \cdot b^{-1} = b \cdot b^{-1} = 1.$$

c)

$$a \cdot 0 = a \cdot (0 + 0) = a \cdot 0 + a \cdot 0.$$

Adding $-(a \cdot 0)$ to both sides, and using associativity, gives $0 = a \cdot 0$. ■

Definition 2.3.3 (The Order Properties of $\mathbb{R}$)

There is a nonempty subset P of $\mathbb{R}$, called the set of **positive real numbers**, that satisfies the following properties.

O1. Trichotomy law. For every $a \in \mathbb{R}$, exactly one of the following holds:

$$a = 0, \quad a \in P, \quad -a \in P.$$

O2. Closure under addition. If $a, b \in P$, then $a + b \in P$.

O3. Closure under multiplication. If $a, b \in P$, then $a \cdot b \in P$.

Remark

O1 (trichotomy) is what makes P usable for case-splitting arguments: given any real number, we may always split into the three exhaustive, mutually exclusive cases $a = 0$, $a \in P$, $-a \in P$, and handle each case separately. O2 and O3 then guarantee that this splitting behaves well under the field operations, so that positivity is preserved by the algebra we already know how to do. This is exactly the pattern used below in the proof that $a \neq 0 \implies a^2 > 0$.

Definition 2.3.4 (Order relation)

Let $a, b \in \mathbb{R}$.

- (i) If $a - b \in P$, then we write $a > b$ or $b < a$.
- (ii) If $a - b \in P \cup \{0\}$, then we write $a \geq b$ or $b \leq a$.

Remark

This definition reduces every order statement to a statement about membership in P : to show $a > b$, it suffices to show $a - b \in P$. All order manipulations later in the course (adding the same quantity to both sides, multiplying by a positive quantity, etc.) ultimately trace back to this definition together with O1–O3.

Remark

$\mathbb{Q}$ is also ordered: the positive rationals $P \cap \mathbb{Q}$ satisfy O1–O3. So, just as with the field axioms, being an ordered field still does not distinguish $\mathbb{R}$ from $\mathbb{Q}$ – we are still missing something. That missing ingredient is completeness, which $\mathbb{Q}$ does *not* have (e.g. $\{q \in \mathbb{Q} : q^2 < 2\}$ is bounded above in $\mathbb{Q}$ but has no least upper bound in $\mathbb{Q}$). This is precisely why completeness must be singled out as a separate axiom rather than derived from the field and order axioms.

Definition 2.3.5 (Completeness Axiom)

If $A \subseteq \mathbb{R}$ is nonempty and bounded above, then A has a least upper bound in $\mathbb{R}$.

Remark

This is only a preview statement: the notions of upper bound and least upper bound (supremum), and the full consequences of this axiom, are developed in §5. For now it suffices to record that this is the axiom that makes $\mathbb{R}$ complete, and hence suitable for limits, while $\mathbb{Q}$ is not.

Example 2.3.6

Prove that if $a \in \mathbb{R}$ and $a \neq 0$, then $a^2 > 0$.

Proof

Since $a \neq 0$, the trichotomy law gives two cases.

Case 1: $a \in P$. By closure under multiplication,

$$a^2 = a \cdot a \in P, \quad \text{so } a^2 > 0.$$

Case 2: $a \notin P$. Since $a \neq 0$, trichotomy rules out $a = 0$, so it must be that $-a \in P$. By closure under multiplication, $(-a)(-a) \in P$. It remains to show that $a^2 = (-a)(-a)$, which we do by starting from $a + (-a) = 0$ and multiplying both sides by $(-a)$:

$$\begin{aligned} a + (-a) = 0 &\implies (-a)(a + (-a)) = (-a) \cdot 0 = 0 \\ &\implies (-a)(a) + (-a)(-a) = 0 \\ &\implies (-a)(-a) = -((-a)a) = -(-a^2) = a^2. \end{aligned}$$

Hence $a^2 = (-a)(-a) \in P$, so $a^2 > 0$. Since the two cases $a \in P$ and $-a \in P$ are exhaustive (trichotomy, together with $a \neq 0$), we conclude $a^2 > 0$ in either case. ■

2.3.1 The Absolute Value

Definition 2.3.7

Given $a \in \mathbb{R}$, we define the **absolute value** $|a|$ by

$$|a| = \begin{cases} a, & a \geq 0, \\ -a, & a < 0. \end{cases}$$

Remark

Geometrically, $|a|$ is the distance from a to 0 on the number line, and more generally $|a - b|$ is the distance between a and b . This interpretation is what makes the theorem and proposition below – especially the triangle inequality – intuitive: they are just algebraic statements about distances.

Theorem 2.3.8 (a) $|ab| = |a||b|$ for all $a, b \in \mathbb{R}$.

(b) $|a|^2 = a^2$ for all $a \in \mathbb{R}$.

(c) For $a \in \mathbb{R}$ and $c > 0$,

$$|a| \leq c \iff -c \leq a \leq c.$$

(d) For every $a \in \mathbb{R}$,

$$-|a| \leq a \leq |a|.$$

Remark (Proof idea)

Parts (a), (b), (d) follow directly by splitting into the cases $a \geq 0$ and $a < 0$ (and similarly for b) and unwinding the definition of $|\cdot|$ in each case – the same case-split strategy as in the $a^2 > 0$ example above. Part (c) is the most important part for later use: it converts a single inequality on $|a|$ into a pair of ordinary inequalities on a , and its proof again splits on the sign of a .

Proof

The definition immediately gives $|-a| = |a|$: if $a \geq 0$, then $-a \leq 0$, and if $a < 0$, then $-a > 0$. It also gives

$$-|a| \leq a \leq |a|,$$

proving (d). In the two cases $a \geq 0$ and $a < 0$, respectively, $|a|^2 = a^2$ is immediate; this proves (b).

For (a), there are four sign combinations. If $a, b \geq 0$, then $ab \geq 0$, so $|ab| = ab = |a||b|$. If $a \geq 0 > b$, then $ab \leq 0$, so $|ab| = -ab = a(-b) = |a||b|$. The case $b \geq 0 > a$ is the same with the roles reversed. Finally, if $a, b < 0$,

then $ab > 0$, and

$$|ab| = ab = (-a)(-b) = |a||b|.$$

Finally let $c > 0$. Suppose first that $|a| \leq c$. If $a \geq 0$, then $|a| = a$, so $a \leq c$; also $-c < 0 \leq a$. If $a < 0$, then $|a| = -a$, so $-a \leq c$, equivalently $-c \leq a$; also $a < 0 < c$. Hence $-c \leq a \leq c$ in either case.

Conversely, suppose $-c \leq a \leq c$. If $a \geq 0$, then $|a| = a \leq c$. If $a < 0$, multiplying $-c \leq a$ by -1 reverses the inequality and gives $-a \leq c$, hence $|a| \leq c$. This proves (c). ■

Remark

Part (c) is the key tool for turning an absolute-value inequality into a two-sided inequality (and vice versa); it will be used constantly in the rest of the course whenever a bound of the form $|x| \leq c$ needs to be unpacked into $-c \leq x \leq c$, or vice versa. The triangle inequality below is the first major illustration of this.

Proposition 2.3.9 (Triangle Inequality)

If $a, b \in \mathbb{R}$, then $|a + b| \leq |a| + |b|$.

Remark (Proof idea)

From part (d) of the preceding theorem we already have the two-sided bounds $-|a| \leq a \leq |a|$ and $-|b| \leq b \leq |b|$. Adding these two chains of inequalities termwise gives a two-sided bound on $a + b$ directly in terms of $|a| + |b|$. This is exactly the situation covered by part (c) of the same theorem, applied with $x = a + b$ and $c = |a| + |b|$: a two-sided bound $-c \leq x \leq c$ converts back into the single inequality $|x| \leq c$, which is precisely the triangle inequality we want.

Proof

Let $a, b \in \mathbb{R}$. By part (d) of the preceding theorem,

$$\begin{aligned} -|a| &\leq a \leq |a|, \\ -|b| &\leq b \leq |b|. \end{aligned}$$

Adding these two chains of inequalities termwise gives

$$\begin{aligned} -|a| - |b| &\leq a + b \leq |a| + |b| \\ \iff -(|a| + |b|) &\leq a + b \leq |a| + |b|. \end{aligned}$$

If $|a| + |b| = 0$, then the displayed bounds force $a + b = 0$, so the claim is immediate. Otherwise $|a| + |b| > 0$, and applying part (c) of the preceding theorem with $x = a + b$ and $c = |a| + |b|$ converts this two-sided bound back into the single inequality

$$|a + b| \leq |a| + |b|. \quad \blacksquare$$

Key mechanism. The bound $-|a| \leq a \leq |a|$ (part (d)) added to its counterpart for b , combined with part (c) ($|x| \leq c \iff -c \leq x \leq c$) applied to $x = a + b$, $c = |a| + |b|$, yields the triangle inequality $|a + b| \leq |a| + |b|$.

Remark

Interpreted via distance, the triangle inequality says that distance is subadditive: the direct distance from 0 to $a + b$ never exceeds the sum of the distances from 0 to a and from 0 to b .

Corollary 2.3.10 (i) $||a| - |b|| \leq |a - b|$.

(ii) $|a - b| \leq |a| + |b|$

Proof of (1)

Let $a, b \in \mathbb{R}$. The idea is to write a as $(a - b) + b$, so that the triangle inequality just proved applies directly to this sum and produces a bound involving $|a - b|$:

$$\begin{aligned} a = (a - b) + b &\implies |a| = |(a - b) + b| \leq |a - b| + |b| \\ &\implies |a| - |b| \leq |a - b|. \end{aligned}$$

Symmetrically, writing b as $(b - a) + a$ and applying the triangle inequality the same way gives

$$\begin{aligned} b = (b - a) + a &\implies |b| = |(b - a) + a| \leq |b - a| + |a| \\ &\implies |b| - |a| \leq |b - a| = |-(a - b)| = |a - b| \\ &\implies |a| - |b| \geq -|a - b|. \end{aligned}$$

Combining the two bounds obtained from these two decompositions,

$$-|a - b| \leq |a| - |b| \leq |a - b|.$$

By part (c) of the absolute value theorem, applied with $x = |a| - |b|$ and $c = |a - b|$, this two-sided bound is equivalent to

$$||a| - |b|| \leq |a - b|.$$

■

Proof of (2)

Let $a, b \in \mathbb{R}$. Writing $a - b$ as $a + (-b)$ reduces this to a direct application of the triangle inequality, using $|-b| = |b|$:

$$|a - b| = |a + (-b)| \leq |a| + |-b| = |a| + |b|.$$

■

2.4 Bounds

Bounds make sense in any ordered set. The special feature of $\mathbb{R}$ is that every nonempty set bounded above has a supremum in $\mathbb{R}$; this need not hold in $\mathbb{Q}$.

Definition 2.4.1 (Bounded Sets)

Let S be a nonempty subset of $\mathbb{R}$. The set S is said to be

- (i) bounded above if there exists $u \in \mathbb{R}$ such that $x \leq u$ for all $x \in S$.
- (ii) bounded below if there exists $w \in \mathbb{R}$ such that $w \leq x$ for all $x \in S$.
- (iii) bounded if it is both bounded above and below.
- (iv) unbounded if it is not bounded.

Definition 2.4.2 (Supremum)

If there exists an upper bound u_0 of S such that whenever u is an upper bound for S we have $u_0 \leq u$, then u_0 is called the least upper bound, or the supremum, of S .

$$u_0 := \sup S.$$

Definition 2.4.3 (Infimum)

If there exists a lower bound w_0 of S such that whenever w is a lower bound for S we have $w_0 \geq w$, then w_0 is called the greatest lower bound, or the infimum, of S .

$$w_0 := \inf S.$$

Remark (Practical recipe)

To show that $u = \sup S$, it suffices to check two things:

- (i) u is an upper bound for S ; and
- (ii) no number smaller than u is an upper bound for S .

Condition (2) is awkward to verify directly, since it quantifies over all real numbers smaller than u . The lemma below rephrases it as an equivalent, checkable condition: for every $\epsilon > 0$ there exists $x \in S$ with $u - \epsilon < x \leq u$, i.e. S contains points arbitrarily close to u from below. The analogous recipe, with inequalities reversed, applies to $\inf S$.

Remark

There can be only one supremum/infimum of a given set S of $\mathbb{R}$.

Idea: Suppose u_1 and u_2 are both suprema of S .

Since u_1 is a supremum and u_2 is an upper bound, we get

$$u_1 \leq u_2.$$

Since u_2 is a supremum and u_1 is an upper bound, we get

$$u_2 \leq u_1.$$

Since $u_1 \leq u_2$ and $u_2 \leq u_1$, we conclude $u_1 = u_2$. Hence the supremum, when it exists, is unique; the same argument (with inequalities reversed) shows the infimum is unique.

Remark (Proof idea)

An upper bound u fails to be the *least* upper bound exactly when some smaller number is still an upper bound, i.e. when there is a "gap" between u and the set S . The ϵ -condition below rules this out: it says that no matter how small a margin $\epsilon > 0$ we allow, S still has a point within that margin of u , so no smaller number can be an upper bound. This turns the definition of supremum (a comparison against *every* upper bound) into a directly checkable statement about S itself.

Lemma 2.4.4

An upper bound u of a nonempty set S in $\mathbb{R}$ is the supremum of S if and only if

$$(\forall \epsilon > 0)(\exists x_\epsilon \in S) \ u - \epsilon < x_\epsilon.$$

Proof

($\Rightarrow$) Suppose $u = \sup S$. Let $\epsilon > 0$. Since

$$u - \epsilon < u,$$

the number $u - \epsilon$ cannot be an upper bound: otherwise it would be an upper bound smaller than u , contradicting that u is the *least* upper bound. Since $u - \epsilon$ is not an upper bound, by the definition of upper bound there is some $x_\epsilon \in S$ such that

$$u - \epsilon < x_\epsilon \leq u.$$

($\Leftarrow$) Conversely, suppose u is an upper bound and that for every $\epsilon > 0$ there exists $x_\epsilon \in S$ with $u - \epsilon < x_\epsilon$. If u were not the least upper bound, there would be another upper bound $w < u$. Choose

$$\epsilon = u - w > 0.$$

The assumed condition then yields some $x_\epsilon \in S$ with

$$w = u - \epsilon < x_\epsilon,$$

contradicting that w is an upper bound (since $x_\epsilon \in S$ exceeds w). Hence no such w exists, and $u = \sup S$. ■

Remark

The ϵ -characterization above is the key practical tool for verifying suprema, and it will be used repeatedly: to confirm $u = \sup S$, exhibit, for each $\epsilon > 0$, a point of S satisfying $u - \epsilon < x \leq u$.

Example 2.4.5

Consider the set $S = \{x \in \mathbb{R} : 0 \leq x < 1\}$.

– $\sup S = 1$.

Proof

Every $x \in S$ satisfies $x < 1$, so 1 is an upper bound. Let $u = \sup S$, which exists by completeness since S is nonempty and bounded above. Because 1 is an upper bound, $u \leq 1$. Suppose $u \neq 1$, so $u < 1$. Choose

$$\epsilon = \frac{1 - u}{2} > 0.$$

Then

$$u < u + \epsilon = \frac{u + 1}{2} < 1.$$

Also $u \geq 0$ since $0 \in S$, so $u + \epsilon \geq 0$. Thus $u + \epsilon \in S$, contradicting that u is an upper bound for S . Therefore $u = 1$, i.e. $\sup S = 1$. ■

– $\inf S = 0$.

Proof

Since $x \geq 0$ for all $x \in S$, 0 is a lower bound. Let $w = \inf S$. Because 0 is a lower bound, $w \geq 0$. Suppose $w \neq 0$, so $w > 0$. But $0 \in S$, and a lower bound w must satisfy $w \leq 0$, a contradiction. Therefore $w = 0$, i.e. $\inf S = 0$. ■

Remark

In this example, $S = [0, 1)$ has $\sup S = 1$ but no maximum, because $1 \notin S$. Its infimum 0 does belong to S , so 0 is also its minimum. A supremum is the *least upper bound*, whether or not it belongs to the set; it is a maximum only when it does belong to the set. Similarly, “bounded above,” “bounded below,” “bounded,” and “unbounded” are properties of the *set* S , whereas $\sup S$ and $\inf S$ (when they exist) are specific *real numbers* associated to S .

2.5 The Completeness Property of $\mathbb{R}$

Axiom (Completeness Property)

Every nonempty set of real numbers that has an upper bound also has a supremum in $\mathbb{R}$.

Corollary 2.5.1

Every nonempty set of real numbers that has a lower bound also has an infimum in $\mathbb{R}$.

Proof

Let $S \subseteq \mathbb{R}$ be nonempty with a lower bound w , that is, $w \leq s$ for all $s \in S$. As a consequence, $-w \geq -s$ for all $s \in S$.

Define $S' = \{-s : s \in S\}$. Then S' is nonempty, and

$$-w \geq -s, \forall s \in S \implies -w \text{ is an upper bound for } S'.$$

By the Completeness Axiom, S' has a supremum. Say

$$u := \sup S' \in \mathbb{R}.$$

We claim that $-u = \inf S$.

- **Lower bound:** Since $u = \sup S' \geq -s$ for all $s \in S$, we get $-u \leq s$ for all $s \in S$. So $-u$ is a lower bound for S .
- **Greatest lower bound:** Let w be any lower bound for S . As shown above, $-w$ is then an upper bound for S' , so $u = \sup S' \leq -w$, which gives $w \leq -u$.

Thus $-u$ is a lower bound for S that is at least as large as every other lower bound, so $-u = \inf S$. ■

Remark

Recall the sup/inf recipe from the previous section: to verify $u = \sup S$, check (i) u is an upper bound for S , and (ii) no number smaller than u is an upper bound for S (equivalently, the ϵ -characterization: for every $\epsilon > 0$ there is $x_\epsilon \in S$ with $u - \epsilon < x_\epsilon$). We will use this recipe repeatedly below, most notably in the Archimedean Property and in the existence of $\sqrt{2}$.

Example 2.5.2

$$S = \left\{ \left(1 + \frac{1}{n}\right)^n : n \in \mathbb{N} \right\}$$

For every $n \in \mathbb{N}$, the binomial theorem gives

$$\left(1 + \frac{1}{n}\right)^n = 1 + \binom{n}{1} \frac{1}{n} + \sum_{k=2}^n \binom{n}{k} \frac{1}{n^k} > 1.$$

Indeed, for $0 \leq k \leq n$,

$$\binom{n}{k} = \frac{n(n-1) \cdots (n-k+1)}{k!} \leq \frac{n^k}{k!}.$$

For $k \geq 2$, we have $k! \geq 2^{k-1}$, because each factor from 2 through k is at least 2. Therefore

$$\left(1 + \frac{1}{n}\right)^n \leq \sum_{k=0}^n \frac{1}{k!} < 1 + 1 + \sum_{k=2}^{\infty} \frac{1}{2^{k-1}} = 3.$$

Thus $1 < (1 + 1/n)^n < 3$ for all $n \in \mathbb{N}$.

2.5.1 The Archimedean Property

We now prove that $\mathbb{N}$ is not bounded above in $\mathbb{R}$. The proof combines two ingredients:

- Completeness of $\mathbb{R}$: this is what lets us form $u = \sup \mathbb{N}$ in the first place. Note carefully that u here is the supremum of $\mathbb{N}$ *regarded as a subset of $\mathbb{R}$* — completeness says nothing about whether u itself is a natural number, and indeed the entire argument below is designed to show that assuming $\mathbb{N}$ is bounded above (so that such a u exists) leads to a contradiction.
- The inductive property of $\mathbb{N}$ (that $\mathbb{N}$ is closed under $+1$): this is the purely arithmetic fact that lets us manufacture a natural number strictly larger than u , which is what breaks the supremum.

Definition 2.5.3 (Archimedean Property)

If $x \in \mathbb{R}$, then there exists $n_x \in \mathbb{N}$ such that $x \leq n_x$.

Remark (Proof idea)

We argue by contradiction: suppose some real number x exceeds every natural number, so that $\mathbb{N}$ is bounded above by x . Completeness then hands us $u = \sup \mathbb{N} \in \mathbb{R}$. The key move is that $u - 1$, being strictly less than u , cannot itself be an upper bound for $\mathbb{N}$ (since u is the *least* upper bound) — this forces some natural number m to lie strictly beyond $u - 1$. But then $m + 1$ is again a natural number (by the inductive property) sitting strictly beyond u , which contradicts u being an upper bound for $\mathbb{N}$ in the first place. So no such x can exist.

Proof

Suppose, for contradiction, that $\exists x \in \mathbb{R}$ such that $x > n$ for all $n \in \mathbb{N}$.

Then x is an upper bound for $\mathbb{N}$. Since $1 \in \mathbb{N}$, $\mathbb{N} \neq \emptyset$, so by the Completeness Axiom, $\mathbb{N}$ has a supremum in $\mathbb{R}$.

Say $u = \sup \mathbb{N}$. Since

$$u - 1 < u,$$

and u is the *least* upper bound of $\mathbb{N}$, no number strictly smaller than u can be an upper bound for $\mathbb{N}$ (an upper bound smaller than the least upper bound would contradict its minimality). Applying this general principle to $u - 1$, we conclude that $u - 1$ is not an upper bound for $\mathbb{N}$. Hence

$$\exists m \in \mathbb{N} \text{ such that } u - 1 < m,$$

which gives

$$u < m + 1.$$

But $m + 1 \in \mathbb{N}$ (since $\mathbb{N}$ is closed under $+1$), and u is an upper bound for $\mathbb{N}$, so $m + 1 \leq u$. This contradicts $u < m + 1$: we have simultaneously $u < m + 1$ and $m + 1 \leq u$.

Therefore no such x exists; that is, for every $x \in \mathbb{R}$ there exists $n_x \in \mathbb{N}$ with $x \leq n_x$. ■

Completeness + closure of $\mathbb{N}$ under $+1 \implies$ Archimedean Property

Remark

The step “ $u-1 < u$, so $u-1$ is not an upper bound” is exactly the ϵ -characterization of the supremum from the previous section, applied with $\epsilon = 1$: if $u = \sup S$, then for every $\epsilon > 0$, $u - \epsilon$ fails to be an upper bound for S , so some element of S exceeds $u - \epsilon$. Here $S = \mathbb{N}$ and $\epsilon = 1$ give the natural number m with $u - 1 < m$.

Corollary 2.5.4

The following are consequences of the Archimedean Property:

(a) If

$$S = \left\{ \frac{1}{n} \mid n \in \mathbb{N} \right\},$$

then

$$\inf S = 0.$$

(b) If $t > 0$, there exists $n_t \in \mathbb{N}$ such that

$$0 < \frac{1}{n_t} < t.$$

(c) If $y > 0$, there exists $n_y \in \mathbb{N}$ such that

$$n_y - 1 \leq y < n_y.$$

Remark (Proof idea)

Part (a) uses the sup/inf recipe: 0 is clearly a lower bound for S , and part (b) shows no larger number can be, since $1/n$ can be made smaller than any prescribed $t > 0$. Part (b) is a direct rescaling of the Archimedean Property: bound $1/t$ above by some natural number m , then $n_t = m + 1$ works. Part (c) is the most subtle: we want the *first* natural number strictly to the right of y , so we collect *all* candidates into a set $A = \{n \in \mathbb{N} : y < n\}$ and take the least element of A — this is a well-ordering argument, not a supremum argument (see the remark following the proof for why).

Proof

- (a) Clearly 0 is a lower bound for S . Let $b > 0$. By part (b), there exists $n \in \mathbb{N}$ such that $1/n < b$. Since $1/n \in S$, this means b is not a lower bound for S . As $b > 0$ was arbitrary, no number greater than 0 is a lower bound for S . Combined with 0 being a lower bound, $\inf S = 0$.
- (b) By the Archimedean Property, choose $m \in \mathbb{N}$ with $1/t \leq m$. Then $n_t = m + 1 \in \mathbb{N}$ satisfies $n_t > m \geq 1/t$, so $n_t > 1/t$, and dividing by $n_t > 0$ gives $0 < 1/n_t < t$.
- (c) We want the least natural number lying strictly to the right of y . To find

it, we collect every natural number with this property into the set

$$A = \{n \in \mathbb{N} : y < n\},$$

and then take the smallest element of A ; this smallest element is automatically the *first* natural number beyond y , which is precisely what we need to pin y between two consecutive integers.

By the Archimedean Property, A is nonempty (some natural number exceeds y). Since $A \subseteq \mathbb{N}$ is nonempty, the Well-Ordering Principle guarantees that A has a least element; let $n_y = \min A$. Then $n_y \in A$, so $y < n_y$.

It remains to show $n_y - 1 \leq y$. Suppose instead that $n_y - 1 > y$. Since $y > 0$ and $n_y \geq 1$ (as $n_y \in \mathbb{N}$), if $n_y = 1$ then $n_y - 1 = 0 > y$ is impossible because $y > 0$; so $n_y \geq 2$ and $n_y - 1 \in \mathbb{N}$. Then $n_y - 1 \in \mathbb{N}$ together with $n_y - 1 > y$ shows $n_y - 1 \in A$. But $n_y - 1 < n_y$, contradicting that n_y is the *least* element of A . Hence $n_y - 1 \leq y$, and combined with $y < n_y$ we obtain

$$n_y - 1 \leq y < n_y.$$

■

Remark

Part (c) is a *minimum* argument, not a supremum argument, and this distinction matters. The set $A = \{n \in \mathbb{N} : y < n\}$ is a subset of $\mathbb{N}$, and in general A is *unbounded above* in $\mathbb{R}$: if y is not itself a boundary case, every sufficiently large natural number also lies in A . So $\sup A$ would not even be a finite real number, and asking for $\sup A$ would be the wrong tool here. What we actually need is the *least* element of A , and the Well-Ordering Principle supplies this regardless of whether A is bounded above — it only requires A to be a nonempty subset of $\mathbb{N}$. This is in direct contrast to the supremum arguments used elsewhere in this section (for $\sup \mathbb{N}$ in the Archimedean Property, and for $\sup S$ in the existence of $\sqrt{2}$ below), where the relevant sets are subsets of $\mathbb{R}$ and boundedness above is essential.

Proposition 2.5.5

There exists a real number x such that $x^2 = 2$.

Remark (Proof idea)

We want to build $\sqrt{2}$ directly as a supremum. Consider

$$S = \{t \in \mathbb{R} : t \geq 0, t^2 < 2\},$$

the set of all nonnegative reals whose square undershoots 2. Intuitively, S consists of all the “candidates” for $\sqrt{2}$ from below, and its supremum $x = \sup S$ should sit exactly at the boundary between squares less than 2 and squares greater than 2 — that is, x^2 should equal 2 precisely. To prove this rigorously, we rule out the two ways x could fail to be $\sqrt{2}$: if $x^2 < 2$, then x is not yet at the boundary and can be nudged slightly to the right while staying in S , contradicting that x is an upper bound; if $x^2 > 2$, then x has overshoot the boundary and can be nudged slightly to the left while remaining an upper bound for S , contradicting that x is the

least upper bound. Both nudges are constructed using the Archimedean Property to choose $n \in \mathbb{N}$ large enough. Note that $x = \sup S$ is a genuine completeness argument: $S \subseteq \mathbb{R}$ is bounded above, and we need the *least upper bound* of a set of reals, not a well-ordering argument as in part (c) above.

Proof

Define

$$S = \{t \in \mathbb{R} : t \geq 0, t^2 < 2\}.$$

We have $1 \in S$, so $S \neq \emptyset$, and 2 is an upper bound for S : if $t \in S$ and $t > 2$, then $t^2 > 4 > 2$, a contradiction. By the Completeness Axiom, S has a supremum in $\mathbb{R}$.

Let $x = \sup S$. Note $x \geq 1 > 0$. We show $x^2 = 2$ by ruling out $x^2 < 2$ and $x^2 > 2$.

Case 1: $x^2 < 2$. We find $n \in \mathbb{N}$ large enough that $x + \frac{1}{n} \in S$, contradicting that x is an upper bound for S .

Step 1 (bound the perturbed square). Since $x \geq 1$ and $n \geq 1$,

$$\begin{aligned} \left(x + \frac{1}{n}\right)^2 &= x^2 + \frac{2x}{n} + \frac{1}{n^2} \\ &\leq x^2 + \frac{2x}{n} + \frac{1}{n} \\ &= x^2 + \frac{2x+1}{n}. \end{aligned}$$

Step 2 (choose n via the Archimedean Property). Since $x^2 < 2$, the quantity $2 - x^2$ is a fixed positive number, so the Archimedean Property lets us choose $n \in \mathbb{N}$ large enough that

$$\frac{2x+1}{n} < 2 - x^2.$$

Step 3 (derive the contradiction). Combining Steps 1 and 2,

$$\left(x + \frac{1}{n}\right)^2 < x^2 + (2 - x^2) = 2,$$

so $x + \frac{1}{n} \in S$. But $x + \frac{1}{n} > x = \sup S$, contradicting that x is an upper bound for S . Hence $x^2 \geq 2$.

$$x^2 < 2 \implies x \text{ can be moved slightly to the right} \\ \text{and remain in } S \implies x \neq \sup S.$$

Case 2: $x^2 > 2$. We find $n \in \mathbb{N}$ large enough that $x - \frac{1}{n}$ is an upper bound for S smaller than x , contradicting that $x = \sup S$.

Step 1 (bound the perturbed square). Since $1/n^2 > 0$,

$$\left(x - \frac{1}{n}\right)^2 = x^2 - \frac{2x}{n} + \frac{1}{n^2} > x^2 - \frac{2x}{n}.$$

Step 2 (choose n via the Archimedean Property). Since $x^2 > 2$, the quantity $x^2 - 2$ is a fixed positive number, so the Archimedean Property lets us choose $n \in \mathbb{N}$ large enough that

$$\frac{2x}{n} < x^2 - 2.$$

Step 3 (derive the contradiction). Combining Steps 1 and 2,

$$\left(x - \frac{1}{n}\right)^2 > x^2 - (x^2 - 2) = 2.$$

So for any $t \in S$, we have $t^2 < 2 < \left(x - \frac{1}{n}\right)^2$. Also $x - 1/n \geq 0$ because $x \geq 1$ and $n \geq 1$. Since $u \mapsto u^2$ is strictly increasing on $[0, \infty)$, this implies $t < x - \frac{1}{n}$. Thus $x - \frac{1}{n}$ is an upper bound for S smaller than $x = \sup S$, contradicting that x is the *least* upper bound. Hence $x^2 \leq 2$.

$x^2 > 2 \implies x$ can be moved slightly to the left and remain an upper bound for $S \implies x \neq \sup S$.

Combining both cases, $x^2 = 2$. ■

2.5.2 Density of $\mathbb{Q}$

Remark

Consider the set

$$T = \{r \in \mathbb{Q} : r^2 < 2\},$$

which mirrors the set S used above to construct $\sqrt{2}$, but restricted to rational numbers. As before, $T \neq \emptyset$ and T is bounded above in $\mathbb{Q}$: for instance $0 \in T$, and every $r \in T$ satisfies $r < 2$.

If T is instead regarded as a subset of $\mathbb{R}$, completeness of $\mathbb{R}$ guarantees that the supremum

$$y = \sup_{\mathbb{R}} T$$

exists. Running exactly the same case-split argument as in the proof above (with T in place of S) shows that $y^2 = 2$, so $y = \sqrt{2}$.

But $\sqrt{2} \notin \mathbb{Q}$, so this candidate supremum y is not itself a rational number. Since y was the *only* possible least upper bound of T in $\mathbb{R}$, no rational number can serve as a least upper bound of T in $\mathbb{Q}$ either — that is, T has no supremum in $\mathbb{Q}$.

We have therefore produced a nonempty subset $T \subset \mathbb{Q}$ that is bounded above in $\mathbb{Q}$ but has no least upper bound in $\mathbb{Q}$. Thus $\mathbb{Q}$ is *not* complete: completeness genuinely depends on $\mathbb{R}$ containing irrational numbers like $\sqrt{2}$ to serve as suprema for sets like T .

Theorem 2.5.6

If x and y are any real numbers with $x < y$, then there exists a rational number $r \in \mathbb{Q}$ such that $x < r < y$.

Remark (Proof idea)

We first treat the case $0 < x < y$. The idea is to build a rational number of the form m/n landing strictly between x and y . First we choose $n \in \mathbb{N}$ large enough (via the Archimedean Property) that $1/n < y - x$; this makes consecutive multiples of $1/n$ closer together than the gap $y - x$, which guarantees that at least one multiple of $1/n$ will fall inside (x, y) . Then we choose m to be the integer just to the right of xn , using the integer-location result (part (c) above) applied to xn . A short computation shows $xn < m < yn$, and dividing by n places $r = m/n$ strictly between x and y . The general case $x \leq 0$ is reduced to the positive case by shifting both x and y up by a large enough integer k , applying the positive case, and shifting back down.

Proof

First suppose that $0 < x < y$. Then $y - x > 0$. By the Archimedean Property, there exists $n \in \mathbb{N}$ such that

$$\frac{1}{n} < y - x.$$

This choice of n ensures that consecutive multiples of $1/n$ are spaced closer together than the gap between x and y , so some multiple of $1/n$ ought to land inside (x, y) ; we now locate it explicitly. Multiplying $1/n < y - x$ by $n > 0$ shows that

$$xn + 1 < yn.$$

We now find a natural number between xn and $xn + 1$. Since $xn > 0$, part (c) of the preceding corollary (with y there replaced by xn) gives some $m \in \mathbb{N}$ with

$$m - 1 \leq xn < m.$$

Equivalently,

$$xn < m \leq xn + 1.$$

This is exactly why part (c) is invoked here: m is, by construction, the first natural number strictly to the right of xn , so $xn < m$ holds automatically. Combining this with $xn + 1 < yn$ obtained above, we have

$$xn < m < yn,$$

where the left inequality is immediate from $m > xn$, and the right inequality follows since $m \leq xn + 1 < yn$. Dividing throughout by $n > 0$, we get

$$x < \frac{m}{n} < y.$$

Since $m \in \mathbb{Z}$ and $n \in \mathbb{N}$, $m/n \in \mathbb{Q}$. This proves the theorem when $0 < x < y$.

Now suppose that $x \leq 0$. Choose $k \in \mathbb{N}$ such that $-x < k$ (possible by the Archimedean Property). Then adding k to $x < y$ and to $-x < k$ gives

$$0 < x + k < y + k.$$

By the positive case just proved, there exists $q \in \mathbb{Q}$ such that $x + k < q < y + k$. Let $r = q - k$. Since $q \in \mathbb{Q}$ and $k \in \mathbb{N} \subseteq \mathbb{Q}$, we have $r \in \mathbb{Q}$, and subtracting k from each part of $x + k < q < y + k$ gives

$$x < r < y.$$

■

Corollary 2.5.7

If x and y are any real numbers with $x < y$, then there exists an irrational number z such that $x < z < y$.

Remark (Proof idea)

Rescale the interval (x, y) by dividing by $\sqrt{2}$, obtaining (a, b) with $a = x/\sqrt{2}$ and $b = y/\sqrt{2}$. Apply the density of $\mathbb{Q}$ to find a nonzero rational $r \in (a, b)$. Then $z = \sqrt{2}r$ lands back in (x, y) , and z must be irrational: if it were rational, $\sqrt{2} = z/r$ would be a ratio of rationals and hence rational, contradicting the irrationality of $\sqrt{2}$.

Proof

Set $a = x/\sqrt{2}$ and $b = y/\sqrt{2}$. Since $\sqrt{2} > 0$, we have $a < b$. Choose a nonzero rational number r with $a < r < b$: if $b \leq 0$, choose $r \in \mathbb{Q}$ with $a < r < b$; if $a \geq 0$, choose $r \in \mathbb{Q}$ with $a < r < b$; and if $a < 0 < b$, choose $r \in \mathbb{Q}$ with $0 < r < b$. In each case, the density theorem provides such an r , and $r \neq 0$.

Let $z = \sqrt{2}r$. Multiplying $a < r < b$ by $\sqrt{2}$ gives $x < z < y$. If z were rational, then $\sqrt{2} = z/r$ would be rational, a contradiction. Thus z is irrational.

■

Corollary 2.5.8

Let $x, y \in \mathbb{R}$, with $x < y$. Then there exists infinitely many rationals between x and y .

Remark (Proof idea)

Apply the density theorem repeatedly: once we have a rational r_k strictly between x and y , the theorem applied to the (still nonempty) interval (r_k, y) produces a new rational r_{k+1} strictly between r_k and y . Iterating this produces an infinite strictly increasing sequence of distinct rationals, all lying in (x, y) .

Proof

Choose $r_1 \in \mathbb{Q}$ with $x < r_1 < y$. Having chosen $r_k \in \mathbb{Q}$ with $x < r_k < y$, apply

the density theorem to $r_k < y$ to choose $r_{k+1} \in \mathbb{Q}$ such that

$$r_k < r_{k+1} < y.$$

This produces infinitely many distinct rational numbers $r_1, r_2, \dots$ in (x, y) . ■

Are there more rational or irrational numbers? To answer this question, we need to know in what sense.

Definition 2.5.9 (Cardinality)

A function $f : A \rightarrow B$ is a *bijection* if it is both injective (one-to-one) and surjective (onto). We say that A and B have the same *cardinality*, written

$$|A| = |B|,$$

if there exists a bijection $f : A \rightarrow B$. Thus, cardinality records the size of a set up to a bijection.

Definition 2.5.10

A set S is *finite* if $S = \emptyset$ or if there is a bijection

$$\{1, 2, \dots, n\} \rightarrow S$$

for some $n \in \mathbb{N}$. A set S is *countably infinite* if there is a bijection $f : \mathbb{N} \rightarrow S$.

Example 2.5.11 (a) The function $f : \mathbb{N} \rightarrow \{2, 4, 6, \dots\}$ defined by $f(n) = 2n$ is a bijection. Thus the set of even natural numbers has the same cardinality as $\mathbb{N}$.

(b) The function $g : \mathbb{N} \rightarrow \mathbb{Z}$ defined by

$$g(1) = 0, \quad g(2k) = k, \quad g(2k+1) = -k \quad (k \in \mathbb{N})$$

is a bijection. Thus $\mathbb{Z}$ has the same cardinality as $\mathbb{N}$.

Definition 2.5.12

A set S is *countable* if it is finite or countably infinite. A set is *uncountable* if it is not countable.

3 Sequences and Limits

3.1 Limits of Sequences

A sequence of real numbers is a function $X : \mathbb{N} \rightarrow \mathbb{R}$. We write $x_n = X(n)$ for its n th term and denote the sequence by $\{x_n\}_{n=1}^{\infty}$.

Definition 3.1.1

A sequence $\{x_n\}$ in $\mathbb{R}$ converges to $a \in \mathbb{R}$ if, for every $\epsilon > 0$, there is an index $n_\epsilon \in \mathbb{N}$ such that

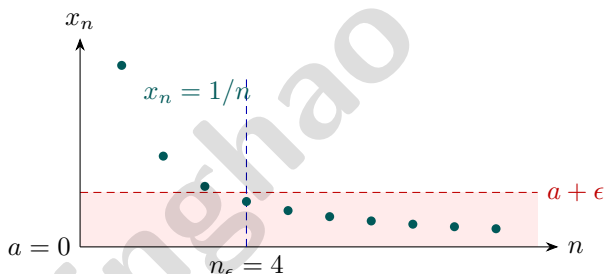
$$n \geq n_\epsilon \implies |x_n - a| < \epsilon.$$

We then write $\lim_{n \rightarrow \infty} x_n = a$.

Remark (Reading the definition)

For every tolerance $\epsilon > 0$, all sufficiently late terms lie in $(a - \epsilon, a + \epsilon)$. The index n_ϵ may depend on ϵ . Once $n \geq n_\epsilon$, the inequality must hold for every later term, even if earlier terms lie outside the interval.

The points below show $x_n = 1/n$. For the illustrated tolerance $\epsilon = 0.3$, every term from $n_\epsilon = 4$ onward lies in the shaded band around $a = 0$.



For $a \in \mathbb{R}$, the ϵ -neighbourhood of a is

$$V_\epsilon(a) = \{x \in \mathbb{R} : |x - a| < \epsilon\} = (a - \epsilon, a + \epsilon).$$

Thus $x_n \rightarrow a$ precisely when, for every $\epsilon > 0$, there is $n_\epsilon \in \mathbb{N}$ such that $x_n \in V_\epsilon(a)$ for all $n \geq n_\epsilon$.

Example 3.1.2

Prove that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

Proof

Let $\epsilon > 0$. By the Archimedean Property, choose $n_\epsilon \in \mathbb{N}$ with $1/n_\epsilon < \epsilon$. For every $n \geq n_\epsilon$,

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} \leq \frac{1}{n_\epsilon} < \epsilon.$$

Hence $1/n \rightarrow 0$. ■

Example 3.1.3

Prove that $\lim_{n \rightarrow \infty} \frac{4n+3}{n+1} = 4$.

Proof

Let $\epsilon > 0$. By the Archimedean Property, choose $n_\epsilon \in \mathbb{N}$ with $1/n_\epsilon < \epsilon$. For every $n \geq n_\epsilon$,

$$\left| \frac{4n+3}{n+1} - 4 \right| = \frac{1}{n+1} \leq \frac{1}{n} \leq \frac{1}{n_\epsilon} < \epsilon.$$

Therefore the sequence converges to 4. ■

Proposition 3.1.4

If $x, y \in \mathbb{R}$ and $|x - y| < \epsilon$ for every $\epsilon > 0$, then $x = y$.

Remark (Proof idea)

If $x \neq y$, their distance is positive. Choose ϵ to be half that distance; the assumed inequality must then fail.

Proof

Suppose $x \neq y$. Then $\epsilon = |x - y|/2 > 0$, but $|x - y| > \epsilon$, contradicting the hypothesis. ■

Theorem 3.1.5

A sequence in $\mathbb{R}$ can have at most one limit.

Remark (Proof idea)

If two limits existed, a sufficiently late term would be close to both. The triangle inequality then makes the two proposed limits arbitrarily close to each other.

Proof

Suppose $x_n \rightarrow a$ and $x_n \rightarrow b$. Given $\epsilon > 0$, choose $n_1, n_2 \in \mathbb{N}$ so that $|x_n - a| < \epsilon/2$ whenever $n \geq n_1$ and $|x_n - b| < \epsilon/2$ whenever $n \geq n_2$. For $n \geq \max\{n_1, n_2\}$, the triangle inequality gives

$$|a - b| \leq |a - x_n| + |x_n - b| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

This holds for every $\epsilon > 0$, so the preceding proposition gives $a = b$. ■

3.1.1 Divergence

A sequence diverges if it does not converge to any real number. For example, consider $\{x_n\} = \{0, 1, 0, 1, \dots\}$, with $x_n = 0$ for odd n and $x_n = 1$ for even n .

Both 0 and 1 keep appearing arbitrarily late, so the terms cannot all remain close to one proposed limit.

Suppose $x_n \rightarrow a$ and take $\epsilon = 1/2$. There is an n_ϵ such that $|x_n - a| < 1/2$ for every $n \geq n_\epsilon$. Choose an odd and an even index beyond n_ϵ . The odd term gives $|a| < 1/2$, hence $-1/2 < a < 1/2$. The even term gives $|1 - a| < 1/2$, hence $1/2 < a < 3/2$. These intervals do not overlap, a contradiction. Thus this sequence diverges.

Definition 3.1.6

A sequence $\{x_n\}$ is bounded if there is an $M \in \mathbb{R}$ such that $|x_n| \leq M$ for every $n \in \mathbb{N}$.

Theorem 3.1.7

Every convergent sequence of real numbers is bounded.

Remark (Proof idea)

Convergence bounds every sufficiently late term near the limit. Only finitely many earlier terms remain, and their absolute values have a maximum.

Proof

Suppose $x_n \rightarrow a$. Taking $\epsilon = 1$, choose $n_1 \in \mathbb{N}$ so that $|x_n - a| < 1$ for all $n \geq n_1$. For these terms, the triangle inequality gives

$$|x_n| \leq |x_n - a| + |a| < 1 + |a|.$$

The remaining terms form a finite set. Thus

$$M = \max\{|x_1|, \dots, |x_{n_1-1}|, 1 + |a|\}$$

bounds every term of the sequence (when $n_1 = 1$, take $M = 1 + |a|$). ■